

Fall 2025, 4100 Problem Set 5, Hayden Fuller

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Let

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & -1 & 3 \\ -6 & 1 & -10 \end{bmatrix}$$

find the LU factorization for A and use it to solve the system

$$\begin{aligned} 2x_1 - x_2 &= 6 \\ 4x_1 - x_2 + 3x_3 &= 1 \\ -6x_1 + x_2 - 10x_3 &= 16 \end{aligned}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -2 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & -4 \end{bmatrix}$$

$$\text{let } b = \begin{bmatrix} 6 \\ 1 \\ 16 \end{bmatrix}, \text{ solve } Ax = b$$

$$Ly = b$$

$$y_1 = 6$$

$$2y_1 + y_2 = 1, y_2 = 1 - 2y_1 = 1 - 12 = -11$$

$$-3y_1 - 2y_2 + y_3 = 16, y_3 = 16 + 3y_1 + 2y_2 = 16 + 18 - 22 = 12$$

$$y = \begin{bmatrix} 6 \\ -11 \\ 12 \end{bmatrix}$$

$$Ux = y$$

$$-4x_3 = 12, x_3 = -3$$

$$x_2 + 3x_3 = -11, x_2 = -11 - 3x_3 = -11 + 9 = -2$$

$$2x_1 - x_2 = 6, 2x_1 + 2 = 6, x_1 = 2$$

$$x = \begin{bmatrix} 2 \\ -2 \\ -3 \end{bmatrix}$$

2. Find the QR factorization of the matrix

$$A = \begin{bmatrix} -1 & 3 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

and use it to find the least squares solution to $A\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$

$$\|a_1\| = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2}$$

$$q_1 = \frac{a_1}{\sqrt{2}} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$r_{12} = q_1^T a_2 = \frac{-3}{\sqrt{2}}, \quad u_2 = a_2 - r_{12}q_1 = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} - \frac{-3}{\sqrt{2}} \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ 1 \\ \frac{3}{2} \end{bmatrix}$$

$$\|u_2\| = \sqrt{\frac{9}{4} + 1 + \frac{9}{4}} = \sqrt{\frac{11}{2}}$$

$$q_2 = \frac{u_2}{\|u_2\|} = \frac{\sqrt{2}}{2\sqrt{11}} \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{3\sqrt{2}}{2\sqrt{11}} \\ \frac{\sqrt{2}}{\sqrt{11}} \\ \frac{3\sqrt{2}}{2\sqrt{11}} \end{bmatrix}$$

$$Q = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{3\sqrt{2}}{2\sqrt{11}} \\ 0 & \frac{\sqrt{2}}{\sqrt{11}} \\ \frac{1}{\sqrt{2}} & \frac{3\sqrt{2}}{2\sqrt{11}} \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{2} & \frac{-3}{\sqrt{2}} \\ 0 & \sqrt{\frac{11}{2}} \end{bmatrix}$$

$$y = Q^T b$$

$$y_1 = q_1^T b = \frac{-2+2}{\sqrt{2}} = 0$$

$$u_2^T b = \frac{3}{2} * 2 + 1 * 1 + \frac{3}{2} * 2 = 7$$

$$y_2 = q_2^T b = \frac{u_2^T b}{\|u_2\|} = \frac{7}{\sqrt{11/2}}$$

$$Rx = y$$

$$\sqrt{\frac{11}{2}}x_2 = y_2, \quad x_2 = \frac{y_2}{\sqrt{11/2}} = \frac{7}{11/2} = \frac{14}{11}$$

$$\sqrt{2}x_1 - \frac{3}{\sqrt{2}}x_2 = 0, \quad x_1 = \frac{3}{2}x_2 = \frac{21}{11}$$

$$x = \begin{bmatrix} 21/11 \\ 14/11 \end{bmatrix}$$

3. Let $A \in \mathbb{R}^{2 \times 2}$ such that $A^2 - A = 6I$ and $|A - 3I| = |A + 2I| = 0$. Compute the determinant of A without using eigenvalues.

$$\det(A - \lambda I) = \lambda^2 - (\text{trace} A)\lambda + \det A$$

$$\det(A - 3I) = \det(A + 2I) = 0$$

$$9 - 3\text{trace} A + \det A = 0$$

$$4 + 2\text{trace} A + \det A = 0$$

$$5 - 5\text{trace} A = 0, \text{trace} A = 1$$

$$9 - 3(1) + \det A = 0$$

$$\det A = -6$$

4. Let $x_1, x_2, x_3 \in \mathbb{R}$ and denote by $\mathbb{P}_2$ the space of all real polynomials of degree at most 2. Let $T \in \mathcal{L}(\mathbb{P}_2, \mathbb{R}^3)$ be the evaluation operator defined by $T : p(x) \rightarrow (p(x_1), p(x_2), p(x_3))$. Find the matrix representation of T with respect to the standard basis, $\{e_i = x^i : i = 0, 1, 2\}$ and show that its determinant is equal to $(x_3 - x_2)(x_3 - x_1)(x_2 - x_1)$.

$$V = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix}$$

$$\det V = \det \begin{bmatrix} 1 & x_1 & x_1^2 \\ 0 & x_2 - x_1 & x_2^2 - x_1^2 \\ 0 & x_3 - x_1 & x_3^2 - x_1^2 \end{bmatrix}$$

$$\det V = (x_2 - x_1)(x_3 - x_1) \det \begin{bmatrix} 1 & x_1 & x_1^2 \\ 0 & 1 & x_2 + x_1 \\ 0 & 1 & x_3 + x_1 \end{bmatrix}$$

$$\det \begin{bmatrix} 1 & x_2 + x_1 \\ 1 & x_3 + x_1 \end{bmatrix} = (x_3 + x_1) - (x_2 + x_1) = x_3 - x_2$$

$$\det V = (x_3 - x_2)(x_3 - x_1)(x_2 - x_1)$$

5. Let A, B, C, D be $n \times n$ matrices such that A is invertible and consider the block matrix
- $$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

- a. Show that the block matrix $E = \begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix}$ reduces M to upper-block-triangular form and deduce that $\det(M) = \det(A)\det(D - CA^{-1}B)$

$$EM = \begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}$$

$$\det(EM) = \det(A)\det(D - CA^{-1}B)$$

$$\det(E) = 1$$

$$\det(M) = \det(E)^{-1}\det(EM) = \det(EM) = \det(A)\det(D - CA^{-1}B)$$

- b. Use the result above to compute the determinant of

$$M = \begin{bmatrix} 1 & 3 & -2 & 1 \\ 4 & 5 & 3 & -2 \\ -1 & 4 & 1 & 3 \\ 2 & -1 & 4 & -5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 3 \\ 4 & 5 \end{bmatrix}, B = \begin{bmatrix} -2 & 1 \\ 3 & -2 \end{bmatrix}$$

$$C = \begin{bmatrix} -1 & 4 \\ 2 & -1 \end{bmatrix}, D = \begin{bmatrix} 1 & 3 \\ 4 & -5 \end{bmatrix}$$

$$\det A = 1 * 5 - 3 * 4 = 5 - 12 = -7$$

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} 5 & -3 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -5/7 & 3/7 \\ 4/7 & -1/7 \end{bmatrix}$$

$$A^{-1}B = \begin{bmatrix} -5/7 & 3/7 \\ 4/7 & -1/7 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 19/7 & -11/7 \\ -11/7 & 6/7 \end{bmatrix}$$

$$CA^{-1}B = \begin{bmatrix} -1 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 19/7 & -11/7 \\ -11/7 & 6/7 \end{bmatrix} = \begin{bmatrix} -9 & 5 \\ 7 & -4 \end{bmatrix}$$

$$D - CA^{-1}B = \begin{bmatrix} 1 & 3 \\ 4 & -5 \end{bmatrix} - \begin{bmatrix} -9 & 5 \\ 7 & -4 \end{bmatrix} = \begin{bmatrix} 10 & -2 \\ -3 & -1 \end{bmatrix}$$

$$\det D - CA^{-1}B = 10 * (-1) - (-2) * (-3) = -10 - 6 = -16$$

$$\det M = \det A \det D - CA^{-1}B = (-7) * (-16) = 112$$

6. Use Cramer's rule to solve the system

$$\begin{array}{rcl} 2x_1 + 3x_2 + x_3 & = & 8 \\ 4x_1 + 7x_2 + 5x_3 & = & 20 \\ -2x_2 + 2x_3 & = & 0 \end{array}$$

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ 0 & -2 & 2 \end{bmatrix}, b = \begin{bmatrix} 8 \\ 20 \\ 2 \end{bmatrix}$$

$$\det A = 48 - 24 - 8 = 16$$

$$\text{let } A_1 = \begin{bmatrix} b & a_2 & a_3 \end{bmatrix}, A_2 = \begin{bmatrix} a_1 & b & a_3 \end{bmatrix}, A_3 = \begin{bmatrix} a_1 & a_2 & b \end{bmatrix}$$

$$\det A_1 = 192 - 120 - 40 = 32$$

$$\det A_2 = 80 - 64 + 0 = 16$$

$$\det A_3 = 80 - 0 - 64 = 16$$

$$x_1 = \frac{\det A_1}{\det A} = 32/16 = 2$$

$$x_2 = \frac{\det A_2}{\det A} = 16/16 = 1$$

$$x_3 = \frac{\det A_3}{\det A} = 16/16 = 1$$

$$(x_1, x_2, x_3) = (2, 1, 1)$$